

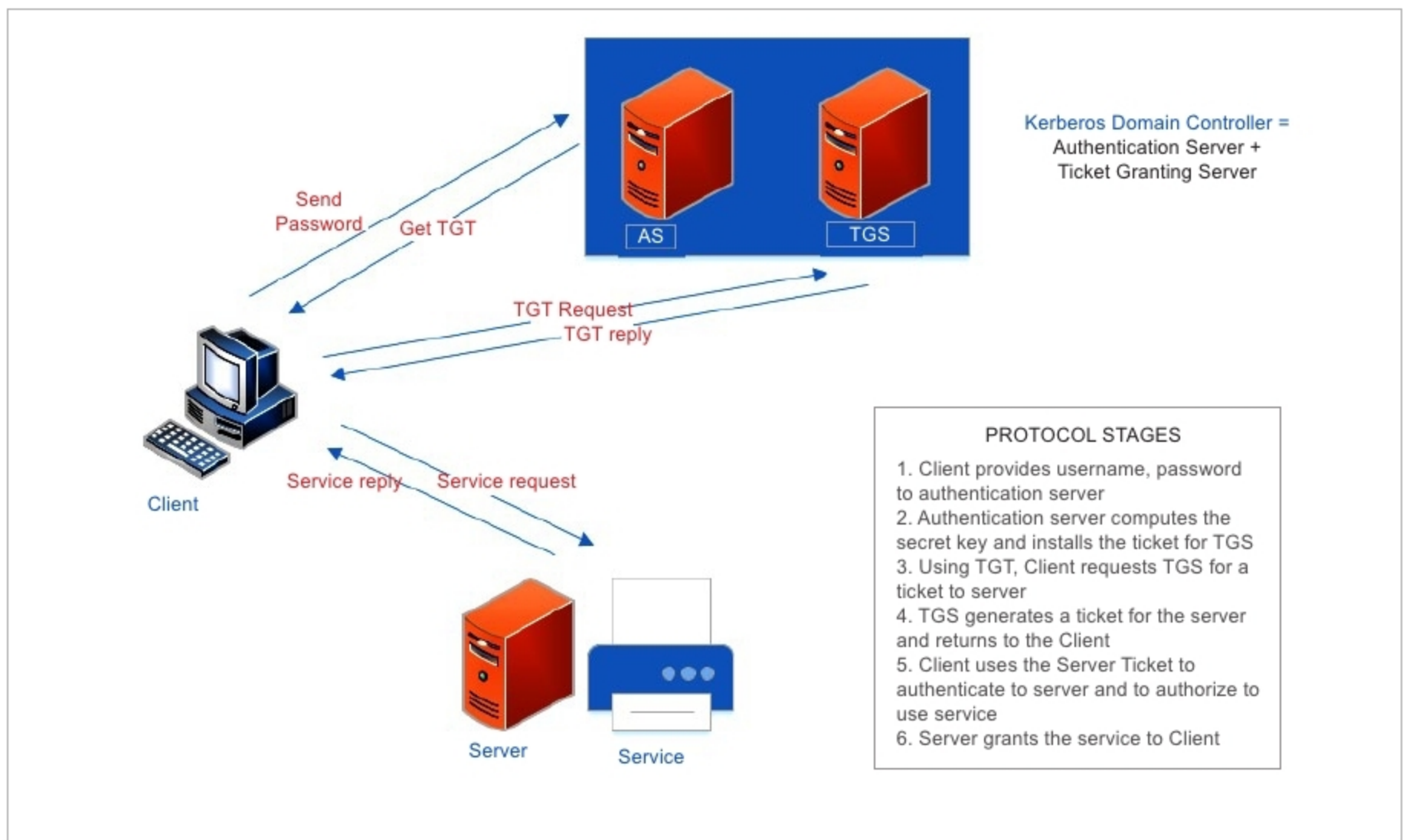


Figure 1: The Kerberos protocol

embedding the IP addresses of clients in the tickets.

- The timestamps used in the tickets restrict the window of time for which messages can be replayed.

Drawbacks of Kerberos

- Kerberos has a single point of failure – the key distribution centre (KDC). So, a continuous availability of KDC has to be ensured.
- Since timestamps are used in Kerberos, it would be required that the clocks of all principals be synchronised.
- If the KDC infrastructure is compromised, it would mean that any attacker can impersonate a user.

Other authentication protocols and a comparison between them

One may come across very many other authentication protocols, some of which also facilitate SSO (single sign-on). A few popular ones

are LDAP or Active Directory, NTLM, OAuth2, SAML, RADIUS and OpenID.

Most of these protocols differ in the way authentication is handled. Some are standards, and some are proprietary. They also differ in use cases. Kerberos, LDAP and NTLM are used more as network protocols in enterprise LANs, while SAML, OAuth2 and OpenID are used with Web applications over HTTP and the RESTful API.

A lot of research goes into formulating cryptographic algorithms to make the Internet more secure. Kerberos has been around for a long time and provides the backbone for many commercial products. It is easy to set it up in Windows or Linux systems. Recent research has provided public key cryptographic extensions to Kerberos authentication, which overcome some of its limitations, paving the way to improved scalability, security and client privacy. **END** 🐧

References

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